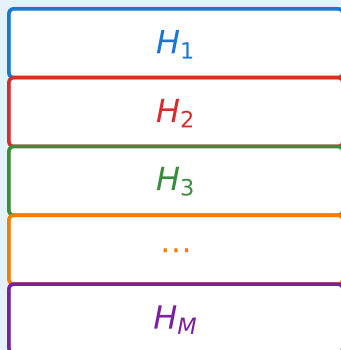


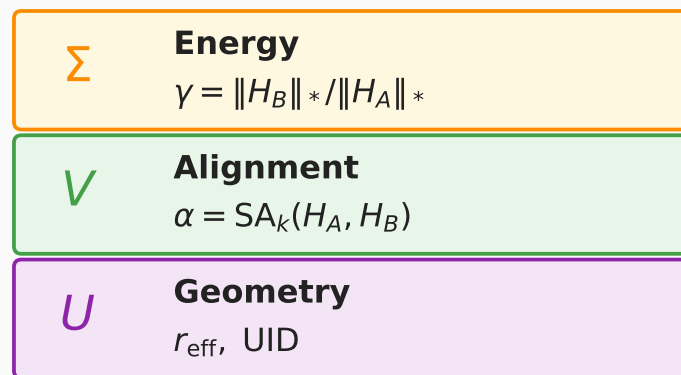
Candidate Pool

M datasets



Three-Channel SVD

per-dataset summary stats



Collapse Model

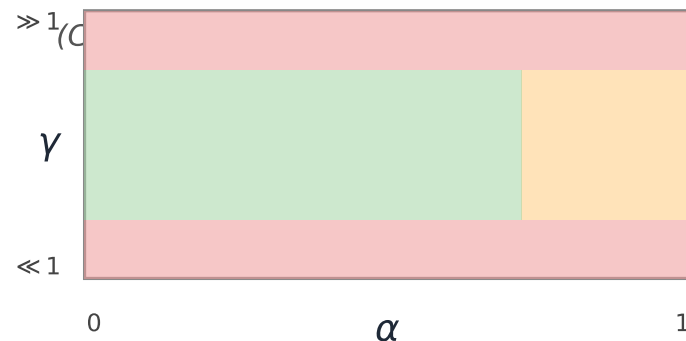
Lemma 1 + Prop. 1 (closed-form)

$$D = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2\alpha}, \quad r^* = \frac{\sqrt{(1 + \gamma^2 + D)/2}}{\sqrt{(1 + \gamma^2 + D)/2} + \sqrt{(1 + \gamma^2 - D)/2}}$$

$$\hat{r}_{\text{eff}} = \exp(H_b(r^*) + r^* \log r_{\text{dom}} + (1 - r^*) \log r_{\text{sub}})$$

superadditivity threshold $\rho_c(\alpha, \gamma)$: $\hat{r}_{\text{eff}} > r_{\text{dom}} \Leftrightarrow r_{\text{dom}}/r_{\text{sub}} < \rho_c$

Three behavior regimes



Superadditive

$\hat{r} > r_{\text{dom}}$
 $\alpha \approx 0, \gamma \approx 1$

Directional collapse

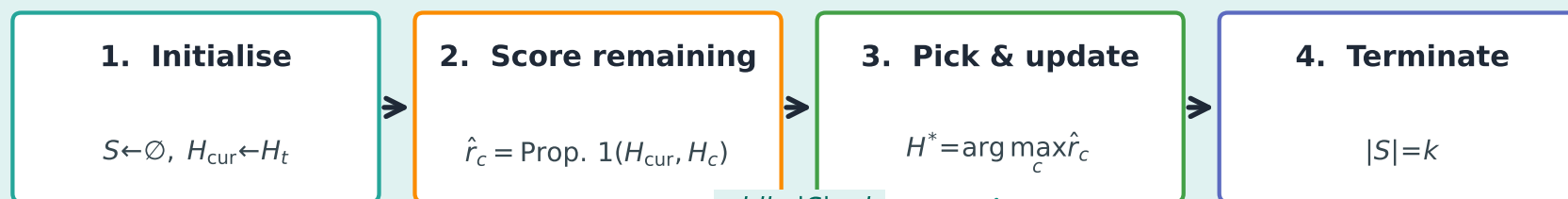
$\hat{r} \rightarrow r_{\text{dom}}$
 $\alpha \rightarrow 1$ (subspace overlap)

Energy collapse

$\hat{r} \rightarrow r_{\text{dom}}$
 $\gamma \gg 1$ or $\gamma \ll 1$

Greedy compression loop

(Algorithm 1; $O(kM)$ predictions over summary stats)



Compressed Pool S

$k \ll M$ datasets

